EcoFlight: Low-Energy Path Finding Through Obstacles for Autonomous Sensing Drones

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***Abstract*—Obstacle avoidance path planning for uncrewed aerial vehicles (UAVs), or drones, is rarely addressed in most flight path planning schemes, despite obstacles being a realistic condition. Obstacle avoidance can also be energy-intensive, making it a critical factor in efficient point-to-point drone flights. To address these gaps, we propose EcoFlight, an energy-efficient pathfinding algorithm that determines the lowest-energy route in 3D space with obstacles. The algorithm models energy consumption based on the drone's propulsion system and flight dynamics. We conduct extensive evaluations, comparing EcoFlight with direct-flight and shortest-distance schemes. The simulation results across various obstacle densities show that EcoFlight consistently finds paths with lower energy consumption than comparable algorithms, particularly in high-density environments. We also demonstrate that a suitable flying speed can further enhance energy savings.**

***Index Terms*—Autonomous drone, A* algorithm, 3D environments, path planning, obstacle avoidance, energy efficiency**

I. Introduction

Drones, or Uncrewed Aerial Vehicles (UAVs), are increasingly used for environmental monitoring, search and rescue, and infrastructure inspection. We have witnessed their increasing deployment in recent disaster response and water quality monitoring in lakes, rivers, and reservoirs [1], [2]. However, their operation time is limited by the battery life. To maximize effectiveness, drones must optimize energy use to extend operational duration while ensuring a safe return to base. While propulsion efficiency contributes to energy savings, planning paths that minimize energy use remains a major challenge—particularly in obstacle-rich environments such as forests and cities.

Previous research has explored several methods for drone navigation, including A*, Dijkstra's algorithm, genetic algorithms, and machine learning-based techniques [3]–[5]. These approaches aim to improve flight efficiency while minimizing computational requirements. However, many face challenges with real-time adjustments, multi-drone coordination, and adapting to dynamic environmental conditions such as wind, weather changes, and obstacles [3], [6].

Although multiple strategies have been proposed for reducing energy consumption, less attention has been given to effectively integrating energy models into path-planning algorithms. To address this gap, we propose EcoFlight, an energy-efficient path planning algorithm based on A* that identifies routes with minimal energy expenditure. Unlike traditional A*, EcoFlight incorporates factors such as altitude changes, flight speed, and acceleration to dynamically select optimal paths. Rather than relying on complex AI training, EcoFlight retains the simplicity and reliability of A* through deterministic estimations, enabling improved efficiency in complex environments.

We evaluated EcoFlight through simulations and compared its performance with alternative path-finding algorithms, including direct path, distance-based A*, rise-and-traverse obstacle avoidance, and obstacle-agnostic direct flight. Results show that EcoFlight consistently outperforms distance-based approaches and delivers greater efficiency, particularly as obstacle density increases.

The remainder of this paper is organized as follows. Section II presents the related work on drone path finding and planning. Section III introduces the proposed Eco-flight path finding algorithm. Section IV compares EcoFlight with state-of-the-art path finding algorithms. Section V concludes the paper.

II. Related Work

Although drone path planning has advanced significantly, several critical challenges remain. Many current methods rely on pre-planned routes and static environmental data, limiting their ability to adapt to dynamic changes such as moving obstacles, shifting wind conditions, or declining battery levels. Traditional path-finding methods like A* and Dijkstra's algorithms perform well in stable environments but struggle in real-time scenarios [7], [8]. Some recent approaches adjust path weights dynamically [9], yet they still fall short when faced with unexpected situations.

Energy estimation presents another challenge. Most models assume that energy consumption depends solely on distance traveled [8], overlooking factors such as acceleration, deceleration, and altitude changes. Some

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studies incorporate variables like wind resistance and payload [10], but these comprehensive models often require substantial computing power, limiting their practicality for real-time applications.

Artificial Intelligence (AI) is also being explored to enhance drone navigation. Techniques such as reinforcement learning and deep learning allow drones to learn from experience and optimize flight paths over time [4]. While promising, these methods require extensive training and high computational resources. Hybrid approaches that combine traditional path planning with AI-based learning [4] may balance efficiency and adaptability, but they still require further testing across diverse environments.

A major barrier to progress is the lack of a standardized framework for evaluating and comparing path-planning methods for moderately long flights between a single origin and destination, and particularly one with obstacles. Realistic flights may encounter obstacles that must be circumvented, yet research often relies on assuming flights on straight lines, with simple flight and energy models. However, efficient flights may be performed close to the ground, therefore, having obstacles in the flight-path is a real possibility. While some works propose some testing environments [11], a widely accepted standard has yet to emerge.

Addressing these challenges is essential for making drone path planning more realistic, reliable, and efficient. Other considerations are incorporating more detailed energy models [3]. Establishing a standardized evaluation framework, specially one with obstacles in a point-to-point flights, would further increase model accuracy and serve as a step forward to flight automation.

III. EcoFlight Algorithm

The proposed EcoFlight algorithm incorporates energy model in path-planning decisions in 3D space.

A. EcoFlight Algorithm

Ecoflight uses an approximation method to evaluate the difference between the predicted energy cost of a path and the remaining energy required. It then applies A* algorithm to select the most energy-efficient route by analyzing the nodes in the 3D map. The flight region is represented as a 3D topographic map, with nodes spaced 1 meter apart and annotated with their elevation. An energy-based scaling factor prioritizes routes that minimize consumption: vertical movement costs are calculated from the drone's mass and gravitational force, with higher penalties for climbing, while horizontal movement costs account for resistance factors such as wind or drag. This approach enables drones to select paths with minimal energy usage, making it well-suited for autonomous operation.

Algorithm 1 presents the pseudocode for EcoFlight. Initially, EcoFlight creates two collections: OpenSet, containing nodes that have been discovered but not yet fully evaluated, and ClosedSet, containing nodes already processed. The algorithm begins by setting g(start)=0 (the energy cost from the start to itself) and computing f(start) = g(start) + h(start, goal), where h estimates the remaining cost to the destination. During execution, the algorithm repeatedly selects the node in OpenSet with the lowest f-value, identifying it as the most promising candidate for an energy-efficient route. If the current node is the goal, the algorithm reconstructs the path by following the parent pointers recorded during traversal and returns this path. Otherwise, the current node is removed from OpenSet, added to ClosedSet, and its neighbors are evaluated.

If a neighbor is already in ClosedSet, it is skipped. For each remaining neighbor, a tentative cost is calculated via the current node. If this cost is lower than any previously recorded cost, or if the neighbor has not yet been discovered, its cost values are updated:g(neighbor) is set to tentative cost, f(neighbor) is represented as g(neighbor)+h(neighbor, goal). The parent pointer is set to the current node. If the neighbor is not in OpenSet, it is added for future evaluation. This process repeats until the goal is reached, in which case the path is reconstructed, or until OpenSet is empty, indicating that no valid path exists.

B. Rise and Traverse Algorithm

The Rise and Traverse algorithm serves as a comparative baseline for evaluating EcoFlight's performance. Unlike EcoFlight, which dynamically searches for energy-efficient paths using heuristic cost functions, Rise and Traverse uses a segmented approach with two distinct phases. The first phase is a vertical ascent to match the drone's altitude with the final destination's z-coordinate. The second phase is a horizontal traversal to the destination. If an obstacle is encountered, an A* search is initiated to find an alternate route, ensuring feasibility while maintaining a preference for upward movement. While computationally simpler, this method lacks dynamic energy optimization and often produces suboptimal paths as compared to EcoFlight. Specifically, it does not incorporate heuristic evaluations for the flight, which can lead to increased energy consumption, particularly in scenarios where diagonal traveling would be more efficient.

C. Energy Model

The energy model incorporates physical parameters such as drone mass, gravitational forces, and aerodynamic drag to calculate energy expenditure along each segment of a flight path using a force vector-based approach.

The energy consumption model in the simulator uses gravitational and kinematics equations that describe the energy consumption on drone flight dynamics.

Table I shows the variables of the physical parameters used to model the energy expenditure estimation.

Algorithm 1 EcoFlight Algorithm

Input: Start node $start$, Goal node $goal$
Output: Optimal path from $start$ to $goal$

OpenSet $\leftarrow \{start\}$
ClosedSet $\leftarrow \emptyset$
$g(start) \leftarrow 0$
$f(start) \leftarrow g(start) + h(start, goal)$
while OpenSet is not empty **do**
  $current \leftarrow$ node in OpenSet with lowest f value
  if $current = goal$ **then**
    return reconstructed path from $current$
  end if
  Remove $current$ from OpenSet
  Add $current$ to ClosedSet
  for all neighbor n of $current$ **do**
    if $n \in$ ClosedSet **then**
      continue
    end if
    tentative_g $\leftarrow g(current) + cost(current, n)$
    if $n \notin$ OpenSet **or** tentative_g $< g(n)$ **then**
      $g(n) \leftarrow$ tentative_g
      $f(n) \leftarrow g(n) + h(n, goal)$
      parent(n) $\leftarrow current$
      if $n \notin$ OpenSet **then**
        Add n to OpenSet
      end if
    end if
  end for
end while
return Failure (No Path Found)

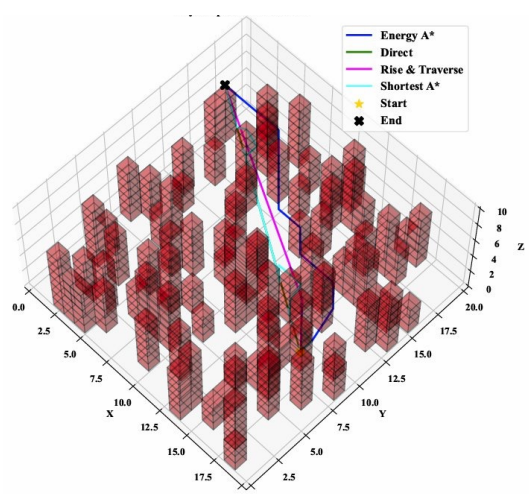


Fig. 1: Sample of an area with obstacles of various heights and the paths followed by the compared path-finding algorithms.

TABLE I: Terms and Physical Parameters.

Parameter	Symbol	Unit
Mass of drone	m	kg
Gravitational acceleration	g	m/s^2
Air density	ρ	kg/m^3
Drag coefficient	C_d	Dimensionless
Cross-sectional area	A	m^2
Cruise speed	v_c	m/s
Acceleration	a	m/s^2

Gravity, thrust and drag [12] forces are defined in (1), (2), and (3), respectively.

$$\vec{F_g} = -mg \tag{1}$$

$$\vec{F_t} = m\vec{a}, \quad \vec{a} = \frac{\vec{v_{\text{new}}} - \vec{v_{\text{old}}}}{\Delta t} \tag{2}$$

$$F_d = \frac{1}{2} C_d A \rho v^2, \quad \vec{F_d} = -F_d \frac{\vec{v}}{|\vec{v}|} \tag{3}$$

When hovering, $\vec{a}$ is an upward acceleration equal to $\vec{g}$.

The *Net Force* is $\vec{F_{\text{net}}} = \vec{F_g} + \vec{F_d} + \vec{F_t}$.

1) Energy Calculations: The energy required is calculated as: work done by net force, hovering energy [13], drag energy [14], and acceleration energy [15] are defined in (4), (5), (6), and (7), respectively.

$$W = \vec{F_{\text{net}}} \cdot \vec{d}, \quad \vec{d} = \vec{p_{\text{new}}} - \vec{p_{\text{old}}}, \tag{4}$$

$$E_h = mgd_t, \tag{5}$$

$$E_d = F_d \cdot d, \tag{6}$$

$$E_a = \frac{1}{2} m (v_{\text{new}}^2 - v_{\text{old}}^2), \tag{7}$$

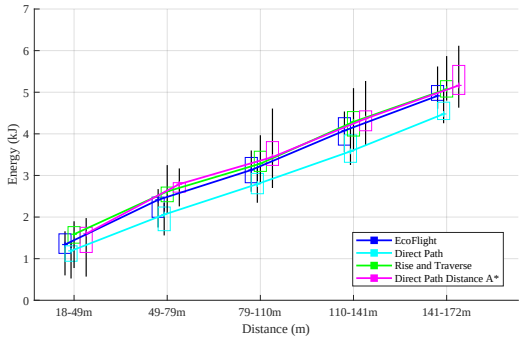
where d_t is the hovering time, as a function of distance and velocity. The energy consumed in each section of the path, or segment, is calculated: Total Energy for Segment i is then:

$$E_{s_i} = E_h + E_d + E_a \tag{8}$$

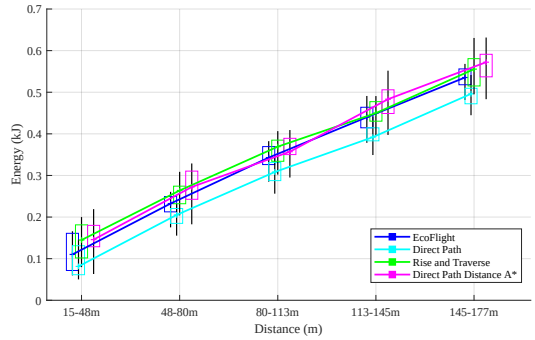
The total energy for a path consisting of N segments [16] is:

$$E_{\text{total}} = \sum_{i=0}^{N-1} E_{s_i} \tag{9}$$

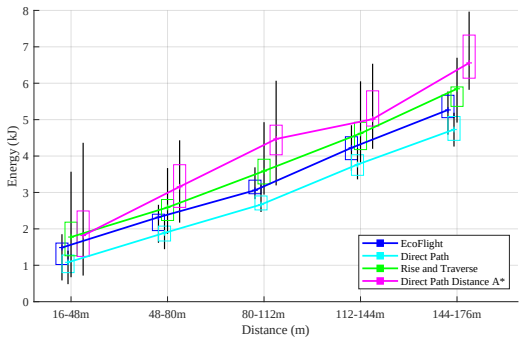
The path energy calculations presented here form the basis for directly comparing different flight paths in terms of energy efficiency. By quantifying the energy expenditure for each path, this approach enables a direct evaluation of EcoFlight's efficiency and effectiveness relative to alternative algorithms. The scheme ensures that path selection prioritizes minimal energy consumption while preserving optimal traversal performance.



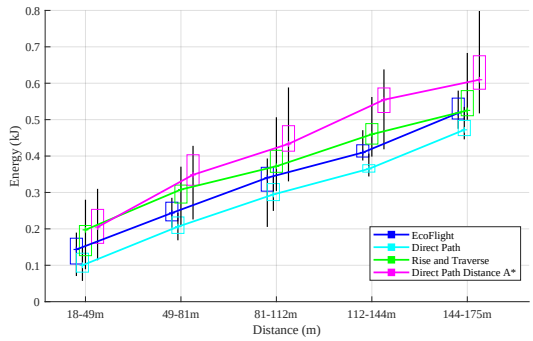
(a) 30% obstacle density at 0.3 m/s.



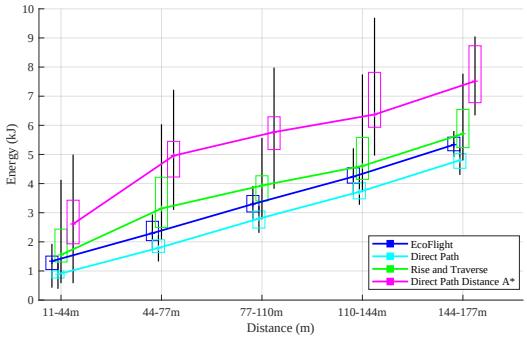
(b) 30% obstacle density at 3 m/s.



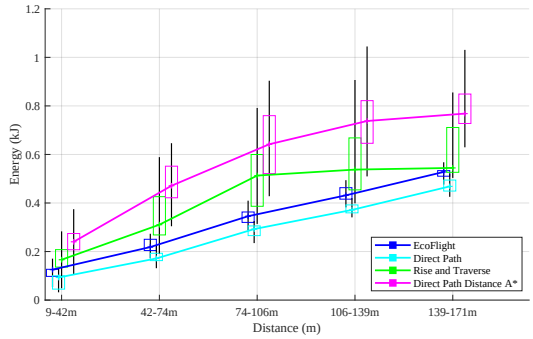
(c) 50% obstacle density at 0.3 m/s.



(d) 50% obstacle density at 3 m/s.



(e) 75% obstacle density at 0.3 m/s.



(f) 75% obstacle density at 3 m/s.

Fig. 2: Comparison of energy consumption of EcoFlight to other path-planning algorithms.

IV. Evaluation

We evaluated EcoFlight's ability to identify energy-efficient paths and compare its energy consumption to that of other path planning methods under various obstacle density environments for calibration and testing.

EcoFlight selects an energy-efficient path that avoids obstacles. Direct Path ignores obstacles entirely, moving in a straight line from start to goal. Direct Path Distance A* follows the shortest path until it encounters an obstacle, at which point it runs a localized and unadjusted A* search to find the shortest detour. Rise and Traverse first moves along the z-axis to match the destination altitude, then proceeds linearly to the goal while avoiding obstacles, similar to Direct Path Distance.

We tested the algorithms under three randomly generated obstacles, with average densities of 30%, 50%, and 75%, and with flights at two different speeds: 0.3 m/s and 3 m/s. Due to the high computational cost, each

configuration was limited to 100 trials. In each trial, the origin and destination were randomly selected within the test area, ensuring the drone maintained a flight altitude at least one unit above any obstacle. Figure 1 shows an example of an area with 30% of the volume with obstacles. The location of the obstacles and their height are randomly selected according to the average density. The drone may fly around or above obstacles through the most energy-efficient path. This figure also shows a glimpse of the possible paths selected by the algorithms used for comparison.

EcoFlight demonstrated remarkable improvements as obstacle density increased. Higher density environments prompted the algorithm to generate smoother and more energy-efficient paths, though at the cost of increased computation time.

As a benchmark, the Direct Path method was the most energy-efficient because this approach ignores the obstacles entirely, cutting straight through them along the shortest possible path. Among the realistic obstacle-aware methods, EcoFlight consistently produced the lowest-energy routes on average. Figures 2a and 2b illustrate the average energy consumption in a simulated environment with 30% obstacle density. With relatively few obstacles, most algorithms produce similar routes and thus exhibit comparable energy usage, forming a tight cluster. However, as obstacle density increases, as seen in Figures 2c and 2d, the need to navigate around obstacles becomes more pronounced, and EcoFlight's advantage becomes clearer. These results also highlight the consistently higher energy cost incurred by Direct Path Distance A* due to its frequent, reactive detours. This trend is further accentuated in Figures 2e and 2f, which correspond to the highest obstacle density tested.

V. Conclusions

In this paper, we proposed EcoFlight, an energy-efficient path planning algorithm that selects energy-efficient flight paths while avoiding obstacles in 3D environment. We evaluated EcoFlight in environments with varying obstacle densities and compared with Direct Path, Rise and Traverse, and Direct Path Distance A* algorithms. Simulation results show that EcoFlight outperforms the other obstacle-aware algorithms in terms of energy consumption. This advantage becomes more significant as obstacle density in the fly zone increases, indicating that some fly maneuvers may be more costly than others and that distance itself may not be proportional to the energy expenditure in the general case. These obstacle scenarios represent spaces of densely populated urban areas with tall buildings. However, the modeling of tree canopies may require a more complex obstacle model to make it realistic.

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